Project Report: Smart Adaptive Streetlight System

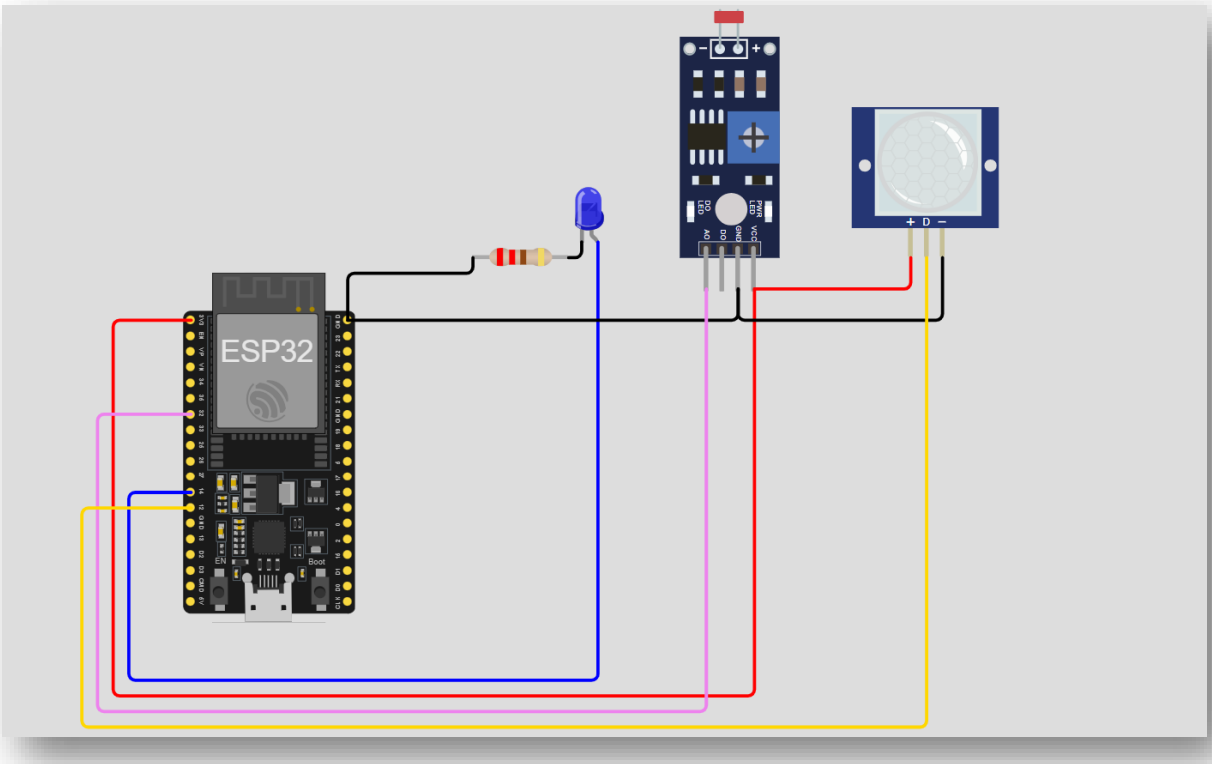
1.Abstract: - Rapid urbanization has led to a massive increase in energy consumption, with street lighting accounting for a significant portion of a city's electricity bill. Traditional streetlights remain at maximum brightness throughout the night, regardless of whether anyone is on the road.

This project introduces a **Smart Streetlight System** built using open-source hardware to drastically reduce energy waste. By combining ambient light sensing, real-time clock synchronization, and motion detection, the system dynamically adjusts its brightness based on actual environmental needs and time of day.

2. Hardware Requirements: - The prototype was built using the following components:

- **ESP32 Microcontroller:** The brain of the project, handling logic, Wi-Fi connectivity for NTP time sync, and PWM (Pulse Width Modulation) output.
- **LDR (Light Dependent Resistor) Module:** Measures ambient sunlight to determine if it's day, dusk, or night.
- **PIR (Passive Infrared) Motion Sensor:** Detects the presence of pedestrians or vehicles.
- **LED & Resistor:** Simulates the actual streetlight.
- **Jumper Wires & Breadboard:** For establishing circuit connections.

3. Circuit Diagram: -



4. Methodology & Working Principle: - The system logic is divided into three main operational phases to maximize power efficiency:

A. Ambient Light Control (Dusk to Dawn): The ESP32 continuously monitors the LDR values. As the sun sets and ambient light drops (LDR value < 3000), the LED gradually turns on. If it becomes completely dark, the system scales the brightness up using the map() function.

B. Time-Based Power Saving Mode (Late Night): The ESP32 connects to the local Wi-Fi and fetches the current Indian Standard Time (IST) using an NTP server (pool.ntp.org). Between 10:30 PM and 5:30 AM, traffic is generally at its lowest. During this window, the system automatically drops the baseline LED brightness to 60% (PWM value 153 out of 255) to conserve power.

C. Motion-Triggered Boost: During the late-night power-saving window, the PIR sensor actively scans for infrared signatures. If a vehicle or pedestrian enters the zone, the PIR sensor triggers a HIGH signal. The ESP32 immediately overrides the power-saving mode and boosts the LED to 100% brightness (PWM 255) for safety, returning to 60% once the area is clear.

D. Automated Logging: For monitoring and maintenance, the system logs its current status (Time, LDR value, PIR status, and current brightness) to the Serial Monitor every 30 minutes.

5. Source Code: -

```
#include <WiFi.h>
#include "time.h"

// Configuration
const char* ssid     = "Debabrata1";
const char* password = "Password@123";

#define pirPin 12
#define ledPin 14
#define ldrPin 32

const int freq = 5000;
const int ledChannel = 0;
const int resolution = 8;

const char* ntpServer = "pool.ntp.org";
const long  gmtOffset_sec = 19800;
const int   daylightOffset_sec = 0;

int ldrValue = 0;
int brightness = 0;
unsigned long lastLogTime = 0;
const unsigned long logInterval = 1800000;

void setup() {
  Serial.begin(9600);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) { delay(500); }
  configTime(gmtOffset_sec, daylightOffset_sec, ntpServer);
  pinMode(pirPin, INPUT);
  pinMode(ldrPin, INPUT);
  ledcAttachChannel(ledPin, freq, resolution, ledChannel);
}

void loop() {
  unsigned long currentMillis = millis();
  ldrValue = analogRead(ldrPin);
  int pirValue = digitalRead(pirPin);

  if (ldrValue < 2000) brightness = 0;
  else if(ldrValue < 3000) brightness = map(ldrValue, 2000, 3000, 40, 150);
```

```
  else brightness = map(ldrValue, 3000, 4095, 150, 255);

  struct tm timeinfo;
  if (getLocalTime(&timeinfo)) {
    int hour = timeinfo.tm_hour;
    int min = timeinfo.tm_min;
    bool isLateNight = ((hour == 22 && min >= 30) || (hour >= 23) || (hour <= 4) || (hour == 5 && min
<= 30));
    if (isLateNight && ldrValue >= 2000) {
      brightness = 153;
      if (pirValue == HIGH) brightness = 255;
    }
  }
  ledcWrite(ledPin, brightness);
  delay(500);
}
```

6. Conclusion and Future Scope: - This prototype successfully demonstrates a highly efficient, responsive street lighting system. By implementing this at scale, municipalities can drastically cut down on power consumption while maintaining public safety.

Future upgrades could include:

- Connecting multiple nodes via LoRa or Zigbee to create a mesh network where streetlights can communicate with each other (e.g., one light detecting motion turns on the next three lights ahead).
- Adding an IoT Dashboard (like Thingspeak or Blynk) to visualize the 30-minute logging data in real-time for city planners.
- Implementing a real-time notification system (via SMS, email, or a centralized dashboard) to alert maintenance crews immediately if a streetlight malfunctions or loses power. Additionally, if the PIR sensor detects continuous, unusual motion during late-night hours, it could trigger a security alert for local authorities.